MONASH UNIVERSITY

FIT5147 Data Exploration and Visualization

Semester 1, 2025

FIT5147 Data Visualization Project (DVP)

TITLE:

Bushfire Impacts Dashboard: Soil, Air & Policy Insights

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APPLIED: 06

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1. Introduction

Bushfires continue to be one of the most critical environmental challenges facing Australia, exacerbated by climate change, prolonged droughts, and increasingly unpredictable weather patterns. These events not only devastate native vegetation and biodiversity but also have far-reaching impacts on soil quality, air pollution levels, and community health. As the frequency and intensity of bushfires rise, it becomes increasingly important to understand their cascading effects across ecological and human systems in a comprehensive, data-driven manner.

This project introduces the *Bushfire Impacts Dashboard*—an interactive narrative visualisation platform that unifies geospatial incident data, laboratory soil chemistry results, air quality readings, and predictive modeling outputs. By weaving together static and temporal datasets into a multi-tabbed narrative, the dashboard provides an integrated exploration of bushfire intensity, soil nutrient depletion, pollutant dynamics, and long-term forecasted risks. Each section of the dashboard presents a targeted storyline, using visual metaphors such as maps, streamgraphs, parallel coordinates, and 3D surface plots to guide users through interconnected patterns and insights.

The visualisation aims to support four key themes:
(1) the spatial spread and intensity of fires,
(2) nutrient leaching and soil degradation,
(3) pollutant spikes and air quality deterioration, and
(4) long-term environmental forecasting.

Through user-driven controls like sliders, dropdown filters, and animated plots, viewers can tailor the insights to their region of interest, pollutant type, or temperature band—enhancing both engagement and analytical depth.

The primary audience for this visualisation includes a wide range of stakeholders such as:

- **Environmental policy makers and land management agencies** seeking data to guide resource allocation, restoration planning, and preventive strategies.
- **Public health officials and regulators** interested in pollutant risk detection and air quality monitoring.
- **Academic researchers and data scientists** looking for structured data insights and visual tools to support environmental modeling and climate adaptation studies.

Ultimately, this dashboard transforms complex multi-dimensional datasets into a cohesive and interactive narrative. By doing so, it empowers informed decision-making, strengthens early warning capabilities, and fosters cross-sector collaboration for climate resilience and ecological recovery.

2. Design Process

2.1 Overview of Five Design Sheets

The Five Design Sheets method, as introduced by Roberts et al., structured the ideation process and helped refine complex ideas into a cohesive, actionable dashboard:

- **Sheet 1: Brainstorming** – Identified key themes (soil, fire, pollution, forecasting), user objectives, and visual concepts. Initial sketches incorporated exploratory visuals like flow diagrams, choropleths, and pollutant time trends tailored to the policy and research audience.
- **Sheet 2: Layout Variant A** – Proposed a card-style layout with grid grouping to support side-by-side comparisons and localised insight panels, echoing Tufte's small multiples principle.
- **Sheet 3: Layout Variant B** – Focused on a vertically-scrolling tab model to encourage narrative flow, prioritising progressive complexity.
- **Sheet 4: Realisation Design** – Integrated realistic components: slider filters, dropdowns, sidebar navigation, titles, and natural language insight boxes.
- **Sheet 5: Final Design** – Unified previous ideas into a four-tab system with shared input logic, high-contrast semantic colour use, and explanatory panels for non-technical users.

2.2 Justification of Design Choices

This section reflects a design process grounded in FIT5147 principles and Munzner's **What-Why-How** framework:

- **What**: spatial fire boundaries, temporal pollutant trends, nutrient concentrations, predictive outputs
- **Why**: users compare regions, interpret patterns, derive predictions
- **How**: via choropleth maps, bar/lollipop charts, streamgraphs, surface plots, and Sankey flows

Consistency and Interaction:

- Visual hierarchy is preserved through repeated UI components (sliders, dropdowns, text panels)
- Tab structures support controlled complexity—each tab presents a self-contained narrative
- Interpretation panels follow the same order and design throughout for familiarity

Colour and Perception:

- **Semantic palette** supports intuitive meaning: warm tones for fire, cool for nutrients, distinct pollutant colours
- Applied **pre-attentive processing principles**: colour for differentiation, position for comparisons
- Colourblind-safe palettes selected for inclusivity and accessibility

Typography and Layout:

- Used sans-serif fonts for on-screen readability, matching web standards
- Information organised using a card-based structure: title → context → graph → insights, leveraging Shneiderman's mantra (overview first, zoom/filter, details-on-demand)

Narrative Style:

- Adopted an **exploratory narrative** model: readers navigate tabs, use filters, and generate insights
- Included **contextual cues and hover tooltips** to reinforce story elements and deepen user agency

Audience Considerations:

- Policy makers: require summary insights, thus the dashboard includes interpretation boxes beneath each graph
- Scientists and analysts: benefit from interactive tooltips and multivariate views (e.g., parallel coordinates)

Each choice—be it layout structure, graph encoding, or narrative sequence—was made to ensure data clarity, reduce cognitive load, and support both lay and technical interpretation. The design matured from open-ended sketches into a harmonised user experience that maintains analytical depth while being visually digestible.

3. Implementation

3.1 Technical Implementation

The dashboard was developed using **R Shiny**, an interactive web application framework in R, to unify geospatial, temporal, and environmental data sources into an interactive narrative. The implementation required integration of multiple datasets, including shapefiles (bushfire perimeters), pollutant time series, and nutrient leachate results.

Key libraries used:

- **Visualisation & UI**: `leaflet` (interactive maps), `plotly` (dynamic plots), `ggplot2` (static chart creation), `reactable` (tabular summary), `networkD3` (flow diagrams), `shinyBS` (collapsible panels)
- **Data wrangling & computation**: `dplyr`, `lubridate`, `tidyr`, `scales`, `akima` (interpolation for surface plots), `randomForest` (prediction model)
- **Geospatial processing**: `sf` (reading shapefiles), `st_make_valid()` used to ensure geometries loaded cleanly

External Code Sources and Adaptations:

- Custom gradient fill functions were adapted from open Plotly community examples
- `akima::interp` was used for generating 3D interpolated nutrient surfaces
- Forecasting approaches used textbook random forest regressors with `caret` pipelines

Implementation Challenges:

- **Data complexity**: required merging shapefiles, cleaning inconsistent pollutant records, and interpolating nutrient surfaces across regions
- **Interactivity logic**: each tab needed synchronized filters (e.g., region and nutrient) that conditionally updated charts and insight panels
- **Performance**: choropleths and streamgraphs required optimisations via reactive expressions and `observeEvent()` to prevent sluggish UI

Design vs Implementation Differences: The implemented dashboard closely follows Sheet 5 from the design process. All proposed components—tab structure, filter controls, interpretation panels—were implemented. Minor changes include resizing maps for better visibility and removing one redundant plot per tab to improve load speed. Overall, the implemented system is faithful to the envisioned design but refined to support performance and usability during runtime.

3.2 Interactive Narrative Visualization Implementation

Each tab supports a narrative with progressive complexity:

Introduction Tab:
The Bushfire Impacts Dashboard is an interactive narrative visualization designed to convey how bushfires in Australia influence soil health, air quality, and future environmental risk. Figure 1 shows that this dashboard equips users with tools to explore, interpret, and plan for bushfire recovery. The intended audience includes policy makers, environmental researchers, and land management agencies seeking data-driven insights to support sustainable intervention strategies.

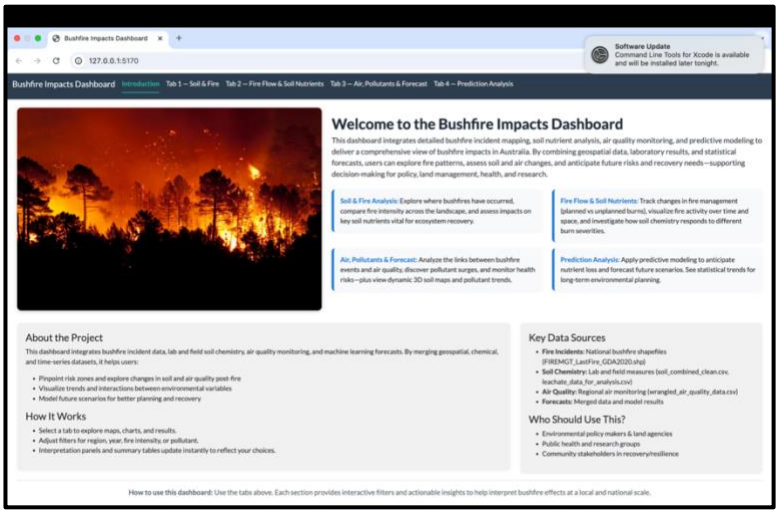

Figure 1: Introduction tab overview presenting the Bushfire Impacts Dashboard, its purpose, data sources, and key user guidance.

Tab 1: Soil & Fire

These visualizations allow users to spatially assess the severity of bushfires and their impact on soil health. Figure 2 compares burn intensities with nutrient centroids to highlight degradation zones, while Figure 3 reveals how nutrient stability shifts under varying fire temperatures, aiding in risk detection and remediation planning.

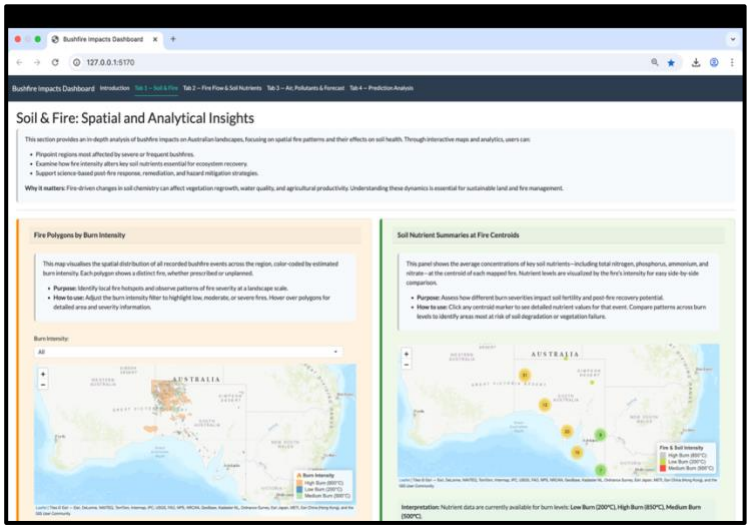

Figure 2: *Choropleth and centroid maps showing bushfire burn intensity and corresponding soil nutrient summaries across regions.*

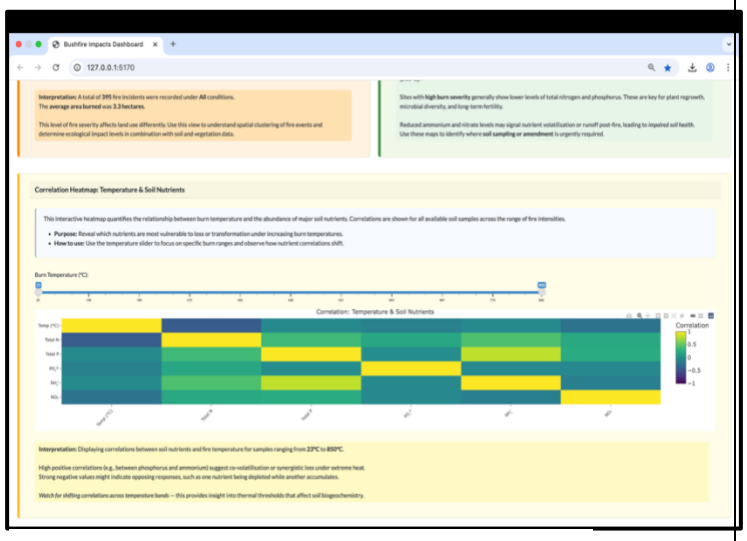

Figure 3: *Interactive correlation heatmap illustrating the relationship between burn temperature and key soil nutrients (e.g., TN, TP, NH_4^+, NO_3^-, PO_4^{3-}) across the 23–850°C range.*

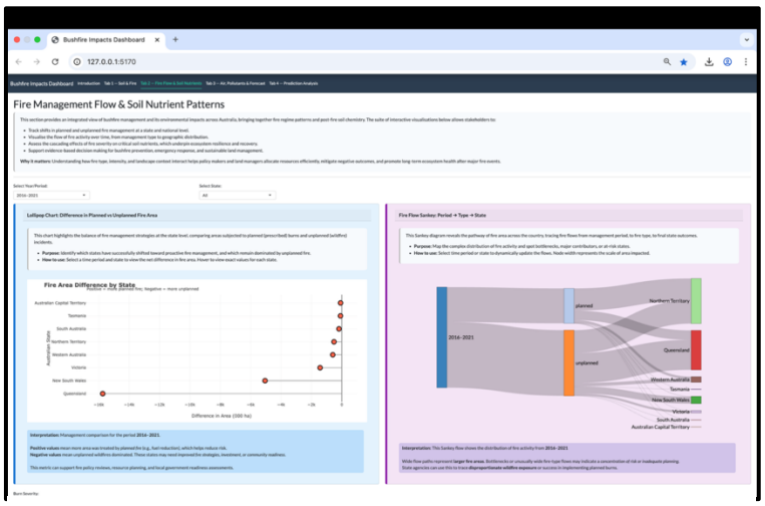

Figure 4: *Comparison of fire management effectiveness across Australian states using lollipop and Sankey diagram for planned and unplanned burns.*

Figure 5: *Visualization of nutrient interdependencies and burn severity effects using parallel coordinates and scatterplot matrix.*

These visuals reveal how fire management strategies vary by state and time, with Figure 4 helping users distinguish between planned and unplanned fire patterns. Figure 5 dives into nutrient responses across burn severities, helping detect interdependencies that inform soil treatment and post-fire recovery strategies.

Tab 3: Air Pollutants & Forecast

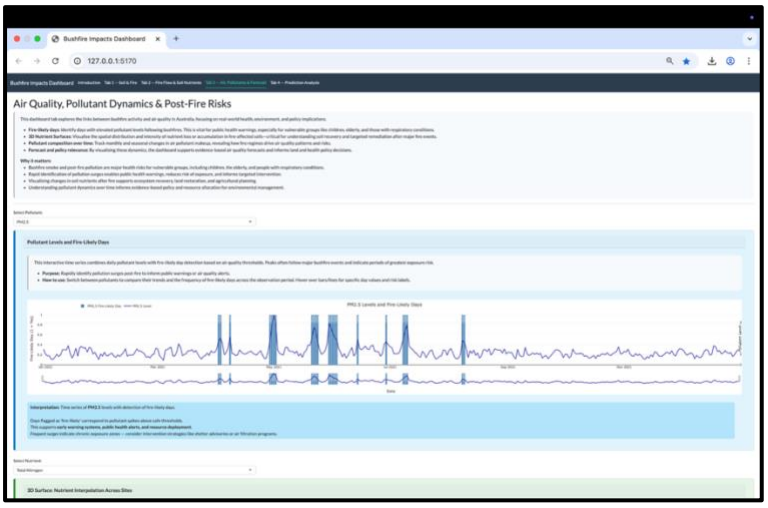

Figure 6: *Time series showing pollutant levels alongside fire-likely day detections to assess air quality risks post-fire.*

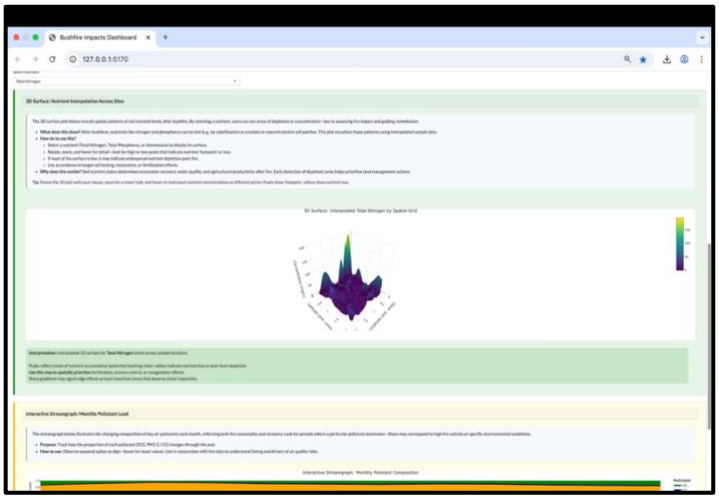

Figure 7: *3D surface plot visualising interpolated Total Nitrogen concentrations across spatial grid to detect post-fire nutrient hotspots and losses.*

These visualizations provide a dynamic view of air pollutant surges following fire events. Figure 6 highlights peak PM2.5 levels and fire-likely days, Figure 7 reveals nutrient surface hotspots post-fire, and Figure 8 tracks seasonal variation in pollutant composition, supporting public health alerts and policy decisions.

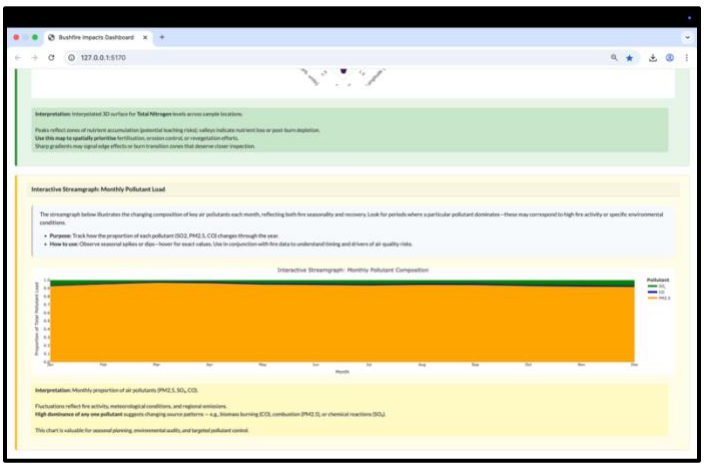

Figure 8: *Streamgraph showing monthly proportions of key air pollutants (PM2.5, SO_2, CO) to visualise seasonal variations and post-fire emission patterns.*

Tab 4: Prediction Analysis

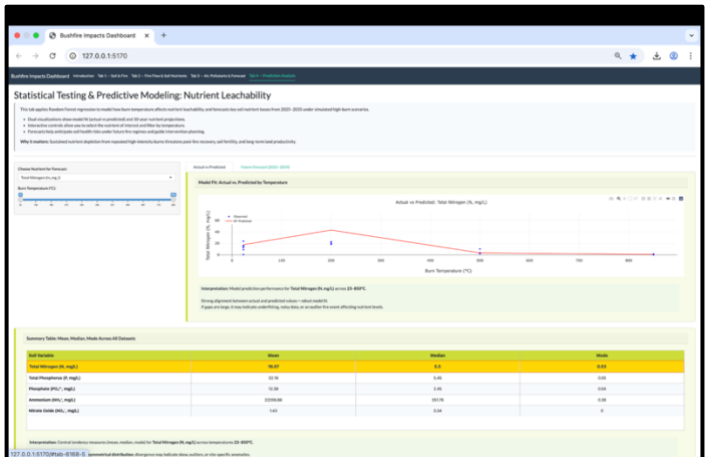

Figure 9: *Random Forest model predictions vs. actual nutrient values across burn temperatures, highlighting fit quality and potential nutrient loss patterns.*

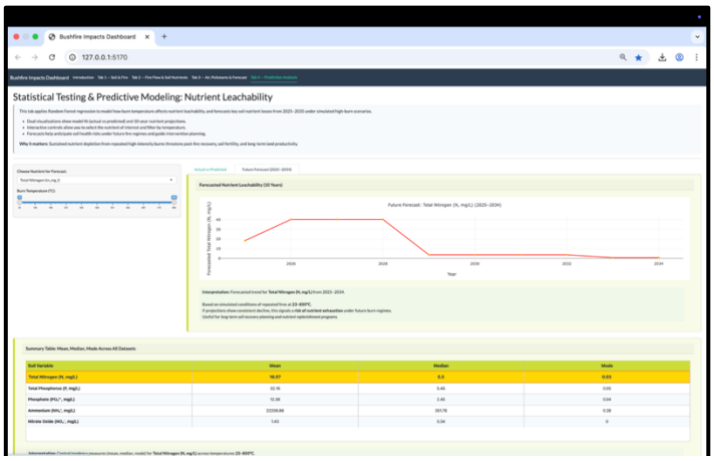

Figure 10: *Forecasted 10-year nutrient leachability trend (2025–2034) under high burn scenarios, illustrating projected declines from Random Forest model outputs.*

These predictive visualizations assess nutrient loss dynamics under varying burn intensities using Random Forest modeling. Figure 9 compares predicted vs. actual nutrient values across temperature bands, validating model fit. Figure 10 projects future leachability trends, indicating declining nutrient availability under continued high-burn conditions—key for long-term soil health planning.

3.3 Using the Implementation

To run the application:

1. Open `ui.R` and `server.R` in RStudio
2. Click **Run App**

Key controls include:

- **Region Selector** (shared across tabs)
- **Year Slider** (Tab 1)
- **Nutrient Dropdown** (Tab 1 & 2)
- **Pollutant Selector** (Tab 3)
- **Forecast Horizon Slider** (Tab 3 & 4)

Interpretation panels below graphs update in real-time. Hover interactions display region/pollutant-specific values. Tabs are arranged logically for progressive exploration. Panels use natural language to summarise patterns.

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4. Conclusion

The Bushfire Impacts Dashboard delivers a cohesive and engaging interactive platform for exploring the multifaceted impacts of bushfires on soil health, air quality, and long-term environmental trends. By synthesising geospatial, chemical, and predictive datasets, it enables users to explore, analyse, and derive meaningful insights for informed decision-making.

Key achievements:

- Created an audience-centric tool grounded in visualisation theory and design frameworks
- Delivered accessible, dynamic visualisations with real-time interactivity and narrative sequencing
- Integrated forecast modelling to support forward-looking environmental assessments

Reflections and Learnings:

Throughout this project, I developed a deep understanding of user-centred design, multi-dimensional data presentation, and the challenges of interactive development. In hindsight, I would have allocated more time to refining the performance of surface plots and implementing advanced ensemble models to improve forecast accuracy.

Future work:

- Integrate real-time APIs (e.g., air quality feeds)
- Expand predictive analysis tab with ensemble learning models
- Conduct usability testing with policymakers and environmental scientists to refine usability and insight delivery

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Roberts, J. C., Headleand, C., & Ritsos, P. D. (2015). Sketching designs using the Five Design-Sheet methodology. *IEEE Transactions on Visualization and Computer Graphics*, *22*(1), 419–428. https://doi.org/10.1109/tvcg.2015.2467271

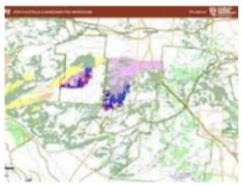
6. Appendix

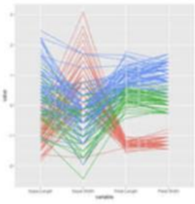
5 SHEET DESIGN

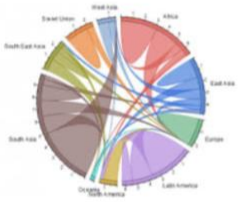
TITLE: DATA VISALIZATION
AUTHOR: RISHITHA JAYADEVA
DATE: 19-05-2025
SHEET: 1
TASK: BRAINSTORM

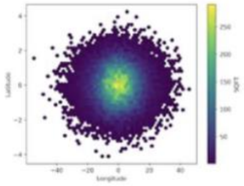
1. IDEAS

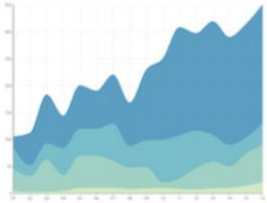
- Fire-scar polygon map
- Year-animated fire map
- Burn-area choropleth (planned vs unplanned)
- Soil nutrient bar chart (TN, TP, NH_4)
- Parallel-coordinates nutrient profiles
- Diverging lollipop chart (planned vs unplanned)
- Sankey diagram (period → type → state → area)
- Stacked pollutant time-series ($PM_{2.5}$, CO, SO_2)
- Monthly pollutant streamgraph
- 3D soil-nutrient surface plot
- Nutrient-forecast line chart (2025–35)
- Chord diagram (state↔type↔TN-bracket)
- Pollutant co-occurrence network graph
- Bivariate choropleth (severity × $\Delta PM_{2.5}$)
- Hexbin heatmap
- Ridge plots of pollutant distributions
- Scatterplot: severity vs pollutant spikes

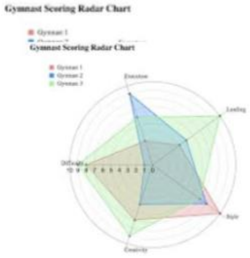












2. FILTER

- Fire-scar polygon map
- Burn-area choropleth (planned vs unplanned)
- Soil nutrient bar chart (TN, TP, NH_4)
- Parallel-coordinates nutrient profiles
- Diverging lollipop chart (planned vs unplanned)
- Sankey diagram (period → type → state → area)
- Stacked pollutant time-series ($PM_{2.5}$, CO, SO_2)
- Monthly pollutant stream graph
- 3D soil-nutrient surface plot
- Nutrient-forecast line chart (2025–35)
- Fire→AQ arc map ($\Delta PM_{2.5}$)
- Pollutant co-occurrence network graph
- Bivariate choropleth (severity × $\Delta PM_{2.5}$)
- Glyph (radar) map

3. CATEGORIZE

- **Spatial:** Fire-scar map, Year-animation,Burn-area choropleth, 3D surface
- **Temporal**: Stacked time-series, Stream graph, Forecast line chart
- **Flow**: Lollipop, Sankey
- **Multivariate**:Parallel-coordinates
- **Composition**: Nutrient bar chart

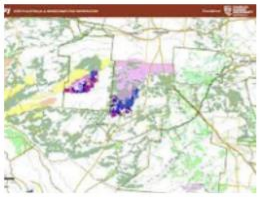
4. COMBINE AND REFINE

- A. Spatial & Soil: Fire-scar map + Year-animated map | Soil nutrient bar | Burn-area choropleth
- B. Flow & Nutrients: Diverging lollipop | Sankey diagram | Parallel-coordinates
- C. Pollutant & Forecast: Stacked time-series | Monthly streamgraph | Nutrient-forecast line

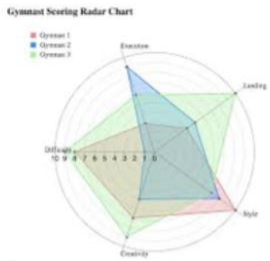
5. QUESTIONS

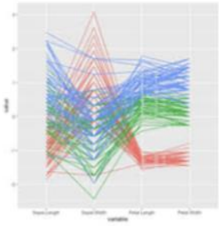
- How does bushfire severity affect soil nutrient leaching?
- How do planned vs. unplanned burns differ by region?
- How do bushfires correlate with urban air-quality pollutant spikes?

LAYOUT



TAB 1





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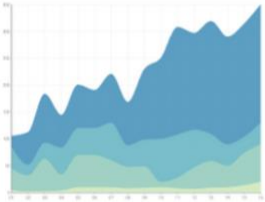
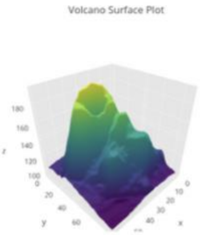


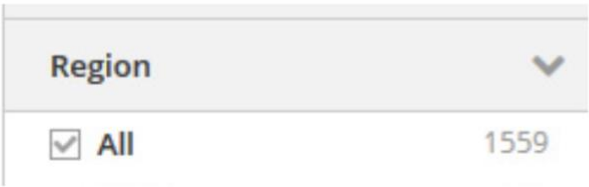


TAB 3

FOCUS

TAB 1 | **TAB2** | **TAB 3**

Clicking a tab header swaps in that tab's three visual panels.



Changing region refreshes chart in the dashboard.

TITLE: DATA VISALIZATION
AUTHOR: RISHITHA JAYADEVA
DATE: 19-05-2025
SHEET: 2
TASK: INITIAL DESIGNS

OPERATIONS

- A region dropdown filters every panel.
- Clicking a tab header switches the entire view to that tab's set of visuals.
- A burn-type toggle recolors the choropleth's bins.
- Hovering any map region shows date, severity, and area burned.
- Hovering any radar axis point reveals its dimension name and score.
- Brushing on the parallel-coordinates plot highlights selected lines and dims others.
- Hovering a parallel-coordinates line displays its full multivariate values.
- A pollutant selector chooses which gas series appear in the streamgraph.
- Brushing the streamgraph's time axis zooms into that interval.
- Rotate and zoom controls on the 3D surface enable close-up inspection.
- Hovering the 3D terrain shows exact nutrient value and spatial coordinates.
- All hovers share a unified tooltip showing panel, region, date/year, metric, and value.

DISCUSSIONS

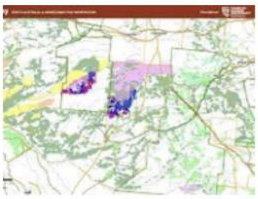
Pros:

- All six panels are visible at once, enabling side-by-side comparison without switching contexts.
- The 2×3 layout provides a balanced, predictable structure that's easy to scan.
- Consistent filters and unified tooltips apply across every view, simplifying interaction.

Cons:

- Six simultaneous charts can still feel busy and may overwhelm some users.
- With only two rows, each panel's size is constrained, limiting detail and readability.
- Multiple legends and hover targets compete for attention in a compact grid.

LAYOUT



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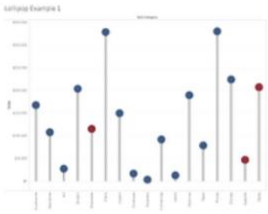
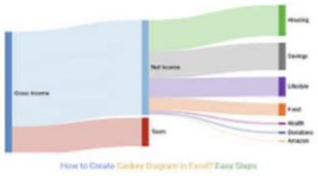
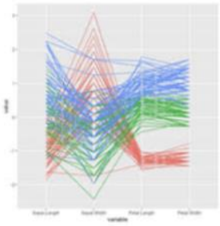
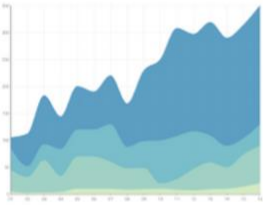
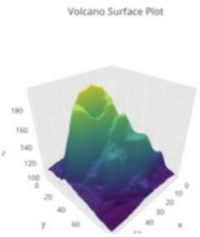






TAB 2

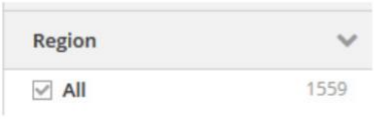




TAB 3

FOCUS

Changing the date refreshes chart in the dashboard.



Changing region refreshes chart in the dashboard.

Picking TN/TP/NH_4 updates the soil bars .

TITLE: DATA VISALIZATION
AUTHOR: RISHITHA JAYADEVA
DATE: 19-05-2025
SHEET: 3
TASK: INITIAL DESIGNS

OPERATIONS

- Date-range slider narrows the time window across all six views.
- A year slider updates the fire-outline layer.
- A baseline toggle flips which burn category serves as the reference across all comparative charts.
- Dragging a node in the flow diagram filters the parallel-coordinates plot to matching records.
- Brushing across the parallel-coordinates axes highlights selected lines and dims the rest.
- The pollutant dropdown now drives both the stacked time-series and the streamgraph.
- Brushing along the time axis zooms both continuous charts in unison.
- Hovering any segment reveals its absolute concentration and its relative monthly proportion.

DISCUSSIONS

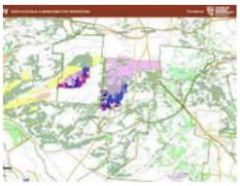
Pros:

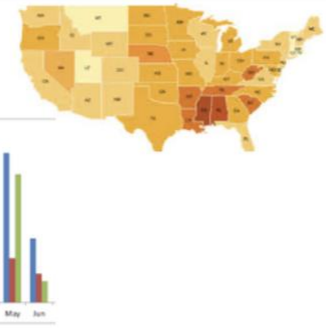
- Tab switching keeps each theme focused and reduces on-screen clutter.
- Linked brushing across both continuous plots provides seamless zooming.
- Hover tooltips showing raw and relative values deliver immediate, dual-context insight.

Cons:

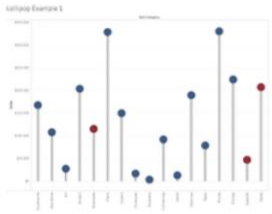
- The baseline toggle adds another layer of complexity for users to learn.
- Brushing both plots simultaneously may overshoot desired focus if scales differ.
- Dense hover interactions can obscure data or overlap in tightly packed visuals.

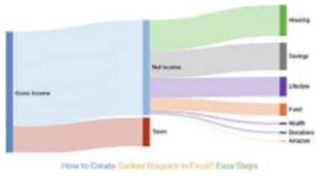
LAYOUT

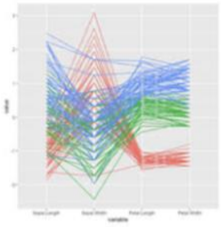




TAB 1







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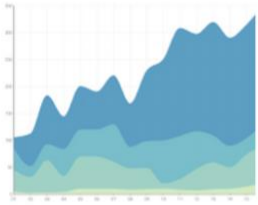
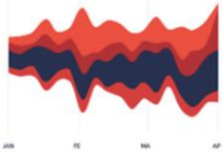
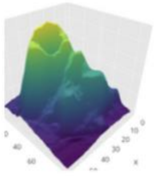
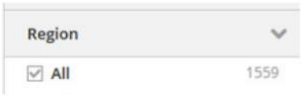






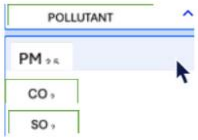
TAB 3

FOCUS



Changing region refreshes chart in the dashboard.

Changing the date refreshes chart in the dashboard.



Selecting $PM_{2.5}$, CO, or SO_2 swaps both time-series and streamgraph views.

Picking TN/TP/NH_4 updates the soil bars .

TITLE: DATA VISALIZATION
AUTHOR: RISHITHA JAYADEVA
DATE: 19-05-2025
SHEET: 4
TASK: REALIZATION DESIGN

OPERATIONS

- The year slider now also updates the soil nutrient bars alongside the fire outlines and choropleth.
- A nutrient dropdown lets you swap among TN, TP, and NH_4 in the bar chart.
- Rotate and zoom controls on the 3D surface enable detailed inspection of nutrient gradients.
- Hovering over the 3D terrain reveals exact concentration values and spatial coordinates.

DISCUSSIONS

Pros:

- Each tab's 2+1 layout groups related charts clearly, reducing visual clutter.
- Shared region and date controls keep interactions consistent across all three views.
- Having the 3D surface in its own panel allows deep spatial inspection without crowding 2D plots.

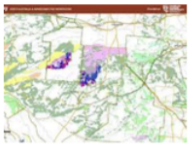
Cons:

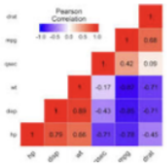
- Switching tabs disrupts cross-theme comparisons and requires extra clicks.
- Learning distinct controls per tab (year slider vs. burn-type toggle vs. forecast horizon) adds complexity.
- The standalone 3D panel may suffer performance lags and has a steeper learning curve.

LAYOUT

INTRODUCTION PAGE

TAB 1





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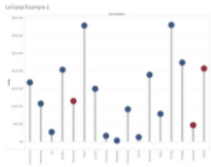
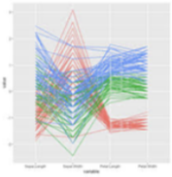
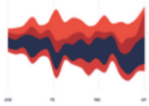
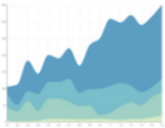
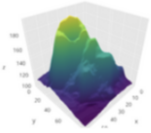




TAB 3







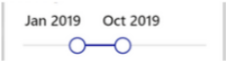
TAB 4

TAB 5

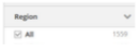
PREDICTION AND STATISTICAL ANALYSIS

FOCUS

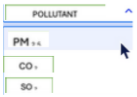
Hovering any mark shows a consistent tooltip with panel, region, date, and value.



Changing the date refreshes chart in the dashboard.



Changing region refreshes chart in the dashboard.



Selecting $PM_{2.5}$, CO, or SO_2 swaps both time-series and streamgraph views.

Picking TN/TP/NH_4 updates the soil bars .

TITLE: DATA VISALIZATION
AUTHOR: RISHITHA JAYADEVA
DATE: 19-05-2025
SHEET: 5
TASK: FINAL DESIGN

OPERATIONS

- The nutrient selector now updates both the soil concentration and the forecast line chart.
- The horizon slider dynamically extends or contracts the forecast period and refreshes the predictions.
- Hovering over forecast points reveals the predicted value alongside its RMSE.
- A unified tooltip displays panel name, region, date/year, metric label, and exact value on hover.

DETAILS

- An interactive web dashboard presenting bushfire impacts through three narrative tabs—Spatial & Soil, Flow & Nutrients, and Pollutant & Forecast—each guiding users through a distinct analytical story.
- Tab-specific widgets (year slider, burn-type toggle, nutrient dropdown, pollutant selector, horizon slider) dynamically update maps, charts, and 3D surfaces to reveal patterns in soil nutrient loss, flow relationships, and air-quality trends.
- Unified tooltips show panel name, region, date/year, metric, and value, ensuring consistent, context-rich insights on every hover.
- Built as a responsive, browser-based application that balances rich interactivity with clear storytelling, making complex environmental data accessible to diverse audiences.

WRANGLING OF DATASETS

#DATASET 1
#wrangling of dataset 1
```{r eval=FALSE}
# 1. Load required libraries

library(readxl)
library(dplyr)
library(janitor)
library(stringr)

# 2. Load Excel data with correct sheet and convert "NR" to NA
df <- read_excel("impact-of-temperatures-on-soil-leachability.xlsx",
                 sheet = "Leachate concentations ",
                 na = "NR")

# 3. Clean column names
df <- janitor::clean_names(df)

# 4. Extract numeric temperature from text like "Unburnt (23)"
temp_col <- grep("temperature.*leach", colnames(df), value = TRUE)
if (length(temp_col) > 0) {
  df$temperature <- gsub("[^0-9]", "", df[[temp_col]])
  df$temperature <- as.numeric(df$temperature)
} else {
  stop("Temperature column not found.")
}

#  5. Coerce character columns to numeric where possible (after NR handling)
df <- df %>%
  mutate(across(where(is.character), ~ suppressWarnings(as.numeric(.)), .names =
"converted_{col}"))

# Replace original numeric columns with converted values (if any)
converted_cols <- grep("^converted_", colnames(df), value = TRUE)
if (length(converted_cols) > 0) {
  for (col in converted_cols) {
    original_col <- gsub("converted_", "", col)
    df[[original_col]] <- df[[col]]
  }
  df <- df[, !colnames(df) %in% converted_cols]  # drop temp columns
}

# 6. Drop completely empty rows
df <- df[rowSums(is.na(df)) != ncol(df), ]
```

```
# 7. Drop columns with >20% missing data
threshold <- 0.2 * nrow(df)
df <- df[, colSums(is.na(df)) < threshold]

# 8. Fill missing values
numeric_cols <- sapply(df, is.numeric)
categorical_cols <- sapply(df, function(x) is.character(x) || is.factor(x))

# Fill numeric with median
df[numeric_cols] <- lapply(df[numeric_cols], function(x) {
  x[is.na(x)] <- median(x, na.rm = TRUE)
  return(x)
})

# Fill categorical with mode
get_mode <- function(x) {
  ux <- unique(x[!is.na(x)])
  ux[which.max(tabulate(match(x, ux)))]
}
df[categorical_cols] <- lapply(df[categorical_cols], function(x) {
  x[is.na(x)] <- get_mode(x)
  return(x)
})

# 9. Remove outliers using IQR
remove_outliers_iqr <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR_val
  upper_bound <- Q3 + 1.5 * IQR_val
  x[x < lower_bound | x > upper_bound] <- NA
  return(x)
}

df[numeric_cols] <- lapply(df[numeric_cols], remove_outliers_iqr)

# Re-impute with median after outlier removal
df[numeric_cols] <- lapply(df[numeric_cols], function(x) {
  x[is.na(x)] <- median(x, na.rm = TRUE)
  return(x)
})

#  Filter valid temperature range
df <- df %>% filter(temperature >= 15, temperature <= 900)
```

```
# 11. Save full cleaned dataset
write.csv(df, "cleaned_leachate_concentrations.csv", row.names = FALSE)

# 12. Select relevant columns for your research questions
# Define actual available columns for analysis
filtered_df <- df %>%
  select(
    temperature,                          # Fire intensity
    tn_mg_l, tp_mg_l, n_ox_mg_l, po4_3_mg_l, nh4_mg_l,  # Nutrients
    zn_mg_l, fe_mg_l, pb_mg_l, cd_mg_l, ni_mg_l, cu_mg_l,
    al_mg_l, as_mg_l, cr_mg_l, co_mg_l, mo_mg_l, mn_mg_l,
    sb_mg_l, se_mg_l, u_mg_l, v_mg_l,              # Metals
    sample_site                           # Site metadata (only available one)
  )

write.csv(filtered_df, "leachate_data_for_analysis.csv", row.names = FALSE)

```

#DATASET 2
#wrangling of dataset 2

```{r eval=FALSE}

# Load required libraries
library(dplyr)
library(readr)

# 📥 Read the cleaned CSVs
table_1 <- read_csv("table_1_data.csv", show_col_types = FALSE)
table_5 <- read_csv("table_5_data.csv", show_col_types = FALSE)

# Standardize columns in table_1
table_1_renamed <- table_1 %>%
  rename(
    state = jurisdiction,
    year_or_period = period
  ) %>%
  mutate(
    year_or_period = as.character(year_or_period),
    data_type = "period_total"
  ) %>%
  select(fire_type, state, year_or_period, area_000ha, data_type)

# Standardize columns in table_5
```

```
table_5_renamed <- table_5 %>%
  rename(
    year_or_period = year
  ) %>%
  mutate(
    year_or_period = as.character(year_or_period),
    data_type = "annual"
  ) %>%
  select(fire_type, state, year_or_period, area_000ha, data_type)

# Combine both datasets
merged_fire_data <- bind_rows(table_1_renamed, table_5_renamed)

# Clean: Remove rows with missing key fields
merged_fire_data_cleaned <- merged_fire_data %>%
  filter(!is.na(fire_type), !is.na(state), !is.na(year_or_period), !is.na(area_000ha))

# Save to single clean CSV file
write.csv(merged_fire_data_cleaned, "merged_fire_data_cleaned.csv", row.names = FALSE)

# Done message
cat("Final merged dataset saved as 'merged_fire_data_cleaned.csv'\n")

```
```

#DATASET - 3
#WRANGLING OF DATASET 3

```{r eval=FALSE}
# Load required libraries
library(readxl)
library(dplyr)
library(janitor)
library(readr)
library(lubridate)
library(stringr)

# Set file path
file_path <- "2021_All_sites_air_quality_hourly_avg_AIR-I-F-V-VH-O-S1-DB-M2-4-0.xlsx"

# 🔍 Get all sheet names
sheet_names <- excel_sheets(file_path)

# Exclude non-data sheets
exclude <- c("Metadata", "Data Info Panel")
data_sheets <- setdiff(sheet_names, exclude)
```
```

```
# Load and clean each sheet, store in a list
cleaned_list <- lapply(data_sheets, function(sheet) {
  df <- read_excel(file_path, sheet = sheet) %>%
    clean_names()
  df$station_id <- sheet
  return(df)
})

# Merge all sheets into one dataframe, padding missing cols
cleaned_air_quality_data <- bind_rows(cleaned_list)

# Optional: Convert column names to lowercase
colnames(cleaned_air_quality_data) <- tolower(colnames(cleaned_air_quality_data))

# Save as CSV
write.csv(cleaned_air_quality_data, "cleaned_air_quality_data.csv", row.names = FALSE)

# Done
cat("Merged and cleaned air quality data saved as 'cleaned_air_quality_data.csv'\n")

# Load the cleaned data
df <- read_csv("cleaned_air_quality_data.csv", show_col_types = FALSE)

# Step 1: Keep only relevant columns
df <- df %>%
  select(datetime_aest, location_name, station_id, bpm2_5, co, so2, dbt)

# Step 2: Remove rows where all pollutant values are NA
df <- df %>%
  filter(if_any(c(bpm2_5, co, so2), ~ !is.na(.)))

# Step 3: Parse datetime
df <- df %>%
  mutate(
    datetime = ymd_hms(datetime_aest),
    date = as.Date(datetime),
    hour = hour(datetime),
    month = month(datetime, label = TRUE),
    year = year(datetime)
  )

#  Step 4: Remove extreme outliers (sensor glitches)
df <- df %>%
  filter(
    is.na(bpm2_5) | bpm2_5 < 500,
```

```
    is.na(co) | co < 20,
    is.na(so2) | so2 < 5
  )

# Step 5: Clean station/location names
df <- df %>%
  mutate(location_name = str_trim(str_to_title(location_name)))

#  Step 6: Add Day/Night label
df <- df %>%
  mutate(period_of_day = if_else(hour >= 6 & hour < 18, "Day", "Night"))

# Step 7: Flag bushfire period
df <- df %>%
  mutate(bushfire_period = if_else(date >= as.Date("2019-06-01") & date <= as.Date("2020-05-
31"), "Yes", "No"))

# (Optional) Step 8: Daily averages summary table
daily_avg <- df %>%
  group_by(location_name, date) %>%
  summarise(
    avg_pm2_5 = mean(bpm2_5, na.rm = TRUE),
    avg_co = mean(co, na.rm = TRUE),
    avg_so2 = mean(so2, na.rm = TRUE),
    avg_temp = mean(dbt, na.rm = TRUE),
    .groups = "drop"
  )

# Save final wrangled dataset and summary
write_csv(df, "wrangled_air_quality_data.csv")
write_csv(daily_avg, "daily_avg_air_quality.csv")

cat("Wrangled dataset saved as 'wrangled_air_quality_data.csv'\n")
cat("Daily average summary saved as 'daily_avg_air_quality.csv'\n")

```
```
#DATASET 4

```{r eval=FALSE}
 # Load required libraries
library(sf)
library(dplyr)
library(lubridate)

# Set working directory
```
```

```
setwd("/Users/rishithajayadeva/Downloads/FIREMGT_LastFire_shp")

# STEP 1: Load the shapefile
fire_data <- st_read("FIREMGT_LastFire_GDA2020.shp")

# STEP 2: Fix invalid geometries
fire_data <- st_make_valid(fire_data)

# STEP 3: Keep only relevant columns
fire_wrangled <- fire_data %>%
  select(
    incident_number = INCIDENTNU,
    incident_type = INCIDENTTY,
    fire_date = FIREDATE,
    financial_year = FINANCIALY,
    fire_year = FIREYEAR,
    season = SEASON,
    date_reliability = DATERELIAB,
    shape_length = SHAPE_Leng,
    shape_area = SHAPE_Area,
    geometry
  )

# STEP 4: Filter valid and reliable records
fire_wrangled <- fire_wrangled %>%
  filter(!is.na(fire_date), date_reliability >= 5) %>%
  mutate(fire_date = as.Date(fire_date))

# STEP 5: Add flags
fire_wrangled <- fire_wrangled %>%
  mutate(
    is_black_summer = fire_year %in% c(2019, 2020),
    is_recent = fire_year >= 2010
  )

# STEP 6: Remove very small polygons
fire_wrangled <- fire_wrangled %>%
  filter(shape_area > 0.0001)

#  STEP 7: Aggregate total burnt area by fire_date
daily_aggregate <- fire_wrangled %>%
  st_drop_geometry() %>%
  group_by(fire_date) %>%
  summarise(total_area_burnt = sum(shape_area, na.rm = TRUE))

# STEP 8: Drop geometry temporarily to perform join
```

```
fire_attributes <- fire_wrangled %>%
  st_drop_geometry() %>%
  left_join(daily_aggregate, by = "fire_date")

# STEP 9: Save final dataset to CSV
write.csv(fire_attributes, "wrangled_fire_data.csv", row.names = FALSE)

cat("Final wrangled dataset saved to 'wrangled_fire_data.csv' with everything needed.\n")

```
```
```